

A Talk on Double Affine Hecke Algebras

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The purpose of this note is to introduce the X - Y duality of double affine Hecke algebras and to illustrate it explicitly in the duality $B_n^{(1)} \leftrightarrow C_n^{(1)}$.

Main Statements

Statement (Root-system and DAHA involution). *For Ion's dual affine root system R^t ,*

$$\alpha_j^t = \frac{A_j}{\sqrt{p}}, \quad \mathring{Q}^t = \frac{1}{\sqrt{p}} M, \quad M^t = \sqrt{p} \mathring{Q}. \quad (1)$$

There is a semilinear algebra isomorphism

$$\phi_{\widetilde{W}} : \mathcal{H}_{\widetilde{W}} \longrightarrow \mathcal{H}_{\widetilde{W}^t} \quad (2)$$

characterized by

$$\phi_{\widetilde{W}}(T_j) = (T_j^t)^{-1}, \quad \phi_{\widetilde{W}}(Y_\mu) = X_{\mu/\sqrt{p}}^t, \quad \phi_{\widetilde{W}}(X_\beta) = Y_{\sqrt{p}\beta}^t, \quad (3)$$

together with

$$\phi_{\widetilde{W}}(q) = (q^t)^{-1}, \quad \phi_{\widetilde{W}}(t_j^{1/2}) = (t_j^t)^{-1/2}. \quad (4)$$

In particular, $(B_n^{(1)})^t = C_n^{(1)}$ and $p = 2$.

Statement (Non-symmetric Macdonald polynomials). *For generic parameters and every $\lambda \in \mathring{Q}$, there is a unique monic triangular common Y -eigenvector*

$$E_\lambda^R = X_\lambda + \sum_{\eta < \lambda} c_{\lambda\eta}^R X_\eta, \quad Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R \quad (\mu \in M). \quad (5)$$

Statement (Non-symmetric evaluation-duality). *Let*

$$X_{\mu/\sqrt{p}}^t(\mathrm{sp}_R(\lambda)) = \gamma_\lambda^R(-\mu), \quad X_\lambda(\mathrm{sp}_{R^t}(\beta)) = \gamma_\beta^{R^t}(-\sqrt{p}\lambda), \quad (6)$$

and normalize E_λ^R and $E_\beta^{R^t}$ at the corresponding vacuum spectral points. Then

$$\boxed{\frac{E_\lambda^R(\mathrm{sp}_{R^t}(\beta))}{E_\lambda^R(\mathrm{sp}_{R^t}(0))} = \frac{E_\beta^{R^t}(\mathrm{sp}_R(\lambda))}{E_\beta^{R^t}(\mathrm{sp}_R(0))}}. \quad (7)$$

All conventions, proofs, recursive constructions, and the explicit $B_n^{(1)}-C_n^{(1)}$ computation are given in the following pages.

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1 Notation and Conventions

We follow Ion's conventions throughout; in particular, $X_\delta = q^{-1}$, the X -index lattice is the finite root lattice $\check{Q}$, and the Y -index lattice is Ion's translation lattice M . The rightmost factor in a product of operators acts first.

Symbol	Meaning
R, R^ι	An affine root system and its Ion dual.
$\alpha_0, \dots, \alpha_n$	The affine simple roots; $\alpha_1, \dots, \alpha_n$ form the finite system.
δ, θ	The null root and the highest finite root, with $\alpha_0 = \delta - \theta$.
$\mathring{Q}$	The finite root lattice, used to index the X -monomials.
M	Ion's translation lattice, generated by $A_j = e_j \alpha_j$, used to index the Y -operators.
$\mathring{W}, W$	The finite and affine Weyl groups.
λ_μ	The translation indexed by $\mu \in M$; at level one it acts by $x \mapsto x + \mu$.
X_β, Y_μ	The two commuting lattice directions of DAHA, indexed by $\beta \in \mathring{Q}$ and $\mu \in M$.
T_j	A Hecke generator corresponding to the simple reflection s_j .
$p = e_s$	Ion's lacing number; $p = 2$ for $B_n^{(1)}$ and $C_n^{(1)}$.
$\phi_{\tilde{W}}$	Ion's semilinear DAHA isomorphism from R to R^ι .
ω_R	The parameter-preserving Fourier map $\phi_{\tilde{W}} \circ \overline{(\cdot)}$.
$\gamma_\lambda^R(\mu)$	The Y_μ -eigenvalue of the non-symmetric Macdonald polynomial E_λ^R .
$\text{sp}_R(\lambda)$	The spectral character on the dual polynomial algebra associated with λ .
$E_\lambda^R, \mathcal{E}_\lambda^R$	The Ion-normalized non-symmetric Macdonald polynomial and its vacuum-normalized version.

When $R = B_n^{(1)}$, the notation E_λ^R is the same as the notation E_λ^I used when emphasizing Ion normalization.

2 Root Data

Let

$$A = (a_{ij})_{0 \leq i, j \leq n} \quad (8)$$

be an irreducible affine Cartan matrix, and let

$$\mathring{A} = (a_{ij})_{1 \leq i, j \leq n} \quad (9)$$

be its finite part. Let

$$\alpha_0, \alpha_1, \dots, \alpha_n \quad (10)$$

be the simple affine roots, and let

$$\mathring{Q} = \bigoplus_{j=1}^n \mathbb{Z} \alpha_j \quad (11)$$

be the finite root lattice. Write

$$\delta = \sum_{j=0}^n a_j \alpha_j, \quad \delta^\vee = \sum_{j=0}^n a_j^\vee \alpha_j^\vee. \quad (12)$$

We exclude type $A_{2n}^{(2)}$. Thus $a_0 = 1$, and

$$\alpha_0 = \delta - \theta, \quad (13)$$

where θ is the highest root of the finite root system.

For $1 \leq j \leq n$, set

$$d_j := \frac{a_j}{a_j^\vee}, \quad e_j := \max\{d_j, a_0^{-1}\}, \quad A_j := e_j \alpha_j, \quad (14)$$

and define

$$M := \bigoplus_{j=1}^n \mathbb{Z} A_j. \quad (15)$$

We regard M as an abelian group of translations $\{\lambda_\mu \mid \mu \in M\}$. The number e_j depends only on the length of α_j . We write e_s and e_l for its values on short and long roots, respectively, and set

$$p := e_s. \quad (16)$$

In the simply laced case, $p = 1$.

We use the symmetric bilinear form

$$(\alpha_j, \alpha_k) := d_j^{-1} a_{jk}, \quad 0 \leq j, k \leq n, \quad (17)$$

and the reflections

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)} \alpha. \quad (18)$$

The finite Weyl group has the Coxeter presentation

$$\mathring{W} = \left\langle s_i \ (1 \leq i \leq n) \ \middle| \ s_i^2 = 1, \ \underbrace{s_i s_j s_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{s_j s_i s_j \cdots}_{m_{ij} \text{ factors}} \ (i \neq j) \right\rangle, \quad (19)$$

where

$$m_{ij} = \begin{cases} 2, & a_{ij} a_{ji} = 0, \\ 3, & a_{ij} a_{ji} = 1, \\ 4, & a_{ij} a_{ji} = 2, \\ 6, & a_{ij} a_{ji} = 3, \\ \infty, & a_{ij} a_{ji} \geq 4. \end{cases} \quad (20)$$

See [Kac83, §3.13]. The affine Weyl group is

$$W = \mathring{W} \ltimes M. \quad (21)$$

3 The Double Affine Hecke Algebra

Let m be the least common denominator of

$$\{(\alpha_j, \Lambda_k) \mid 1 \leq j, k \leq n\}, \quad (22)$$

where Λ_k is the k -th fundamental weight, and let $\mathbb{F}$ be the field of rational functions in $q^{1/m}$ and the $t_j^{1/2}$.

The two lattice algebras are

$$\mathbb{F}[M] = \bigoplus_{\mu \in M} \mathbb{F}Y_\mu, \quad \mathbb{F}[\mathring{Q}] = \bigoplus_{\beta \in \mathring{Q}} \mathbb{F}X_\beta, \quad (23)$$

with

$$Y_\mu Y_\nu = Y_{\mu+\nu}, \quad X_\beta X_\gamma = X_{\beta+\gamma}. \quad (24)$$

The parameters are constant on root-length orbits:

$$d_j = d_k \iff t_j = t_k. \quad (25)$$

3.1 The finite Hecke algebra

The finite Hecke algebra $\mathcal{H}_{\mathring{W}}$ is generated by $T_1, \dots, T_n$, subject to the braid relations corresponding to (19) and the quadratic relations

$$T_j - T_j^{-1} = t_j^{1/2} - t_j^{-1/2}. \quad (26)$$

3.2 The affine Hecke algebra

The affine Hecke algebra $\mathcal{H}_W$ is generated by $\mathcal{H}_{\mathring{W}}$ and $\mathbb{F}[M]$, subject to

$$Y_\mu T_j - T_j Y_{s_j(\mu)} = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{Y_\mu - Y_{s_j(\mu)}}{1 - Y_{A_j}}, \quad (27)$$

for $\mu \in M$ and $1 \leq j \leq n$.

3.3 The double affine Hecke algebra

Definition 3.1. The double affine Hecke algebra $\mathcal{H}_{\tilde{W}}$ is generated by $\mathcal{H}_W$ and $\mathbb{F}[\mathring{Q}]$, subject to

$$T_j X_\beta - X_{s_j(\beta)} T_j = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{X_\beta - X_{s_j(\beta)}}{1 - X_{-\alpha_j}}, \quad (28)$$

for $\beta \in \mathring{Q}$ and $0 \leq j \leq n$, together with

$$X_\delta = q^{-1}. \quad (29)$$

For $j = 0$, the monomial $X_{s_0(\beta)}$ is interpreted using $X_\delta = q^{-1}$.

Thus $\mathcal{H}_{\widetilde{W}}$ contains two commuting lattice algebras:

$$\mathbb{F}[\mathring{Q}] \quad \text{generated by the } X_\beta, \quad (30)$$

and

$$\mathbb{F}[M] \quad \text{generated by the } Y_\mu. \quad (31)$$

The involution below exchanges these two directions.

4 The Root-System and DAHA Involutions

4.1 The root-system involution

Definition 4.1. Define a new affine root system R^ι by

$$\alpha_j^\iota := \frac{A_j}{\sqrt{p}} = \frac{e_j}{\sqrt{p}} \alpha_j, \quad 1 \leq j \leq n, \quad (32)$$

together with

$$\delta^\iota = \delta. \quad (33)$$

The affine simple root α_0^ι is determined by the affine type.

Under the natural identification of the finite root spaces,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{p}} M, \quad M^\iota = \sqrt{p} \mathring{Q}. \quad (34)$$

The involution preserves the affine type in the simply laced cases. In particular,

$$A_n^{(1)\iota} = A_n^{(1)}. \quad (35)$$

In the non-simply-laced classical cases,

$$B_n^{(1)\iota} = C_n^{(1)}, \quad C_n^{(1)\iota} = B_n^{(1)}. \quad (36)$$

On finite roots, ι interchanges short and long roots.

4.2 Ion's semilinear DAHA isomorphism

Let $\mathcal{H}_{\widetilde{W}^\iota}$ be the DAHA associated with R^ι . Its generators are denoted by

$$T_j^\iota, \quad X_\gamma^\iota, \quad Y_\nu^\iota. \quad (37)$$

Theorem 4.2 (Ion's DAHA involution). *The assignments*

$$\phi_{\widetilde{W}}(T_j) = (T_j^\iota)^{-1}, \quad 1 \leq j \leq n, \quad (38)$$

$$\phi_{\widetilde{W}}(Y_\mu) = X_{\mu/\sqrt{p}}^\iota, \quad \mu \in M, \quad (39)$$

$$\phi_{\widetilde{W}}(X_\beta) = Y_{\sqrt{p}\beta}^\iota, \quad \beta \in \mathring{Q}, \quad (40)$$

extend to a semilinear algebra isomorphism

$$\phi_{\tilde{W}} : \mathcal{H}_{\tilde{W}} \longrightarrow \mathcal{H}_{\tilde{W}^\iota}, \quad (41)$$

with

$$\phi_{\tilde{W}}(q) = (q^\iota)^{-1}, \quad \phi_{\tilde{W}}(t_j^{1/2}) = (t_j^\iota)^{-1/2}. \quad (42)$$

Its inverse is $\phi_{\tilde{W}^\iota}$.

This is Ion's Theorems 2.2–2.3 [Ion03, Theorems 2.2–2.3]. Its content is summarized by

$$\boxed{X_\beta \longleftrightarrow Y_{\sqrt{p}\beta}^\iota}. \quad (43)$$

Thus the DAHA involution exchanges the X -lattice and the Y -lattice, while interchanging the root system with its ι -dual.

4.3 The parameter-preserving Fourier map

Let $\overline{(\cdot)}$ denote the standard semilinear bar involution, characterized on the relevant generators by

$$\overline{q} = q^{-1}, \quad \overline{t_j^{1/2}} = t_j^{-1/2}, \quad \overline{T_j} = T_j^{-1}, \quad \overline{X_\beta} = X_{-\beta}, \quad \overline{Y_\mu} = Y_{-\mu}. \quad (44)$$

Define

$$\omega_R := \phi_{\tilde{W}} \circ \overline{(\cdot)}. \quad (45)$$

After identifying the transported parameters, ω_R is parameter-preserving and satisfies

$$\omega_R(T_j) = T_j^\iota, \quad \omega_R(Y_\mu) = X_{-\mu/\sqrt{p}}^\iota, \quad \omega_R(X_\beta) = Y_{-\sqrt{p}\beta}^\iota. \quad (46)$$

Remark 4.3. The sign distinction is important. Ion's semilinear map $\phi_{\tilde{W}}$ sends X_β to $Y_{+\sqrt{p}\beta}^\iota$, whereas the parameter-preserving Fourier map ω_R used in evaluation-duality sends it to $Y_{-\sqrt{p}\beta}^\iota$.

5 Non-symmetric Macdonald Polynomials

The basic polynomial representation is

$$\mathcal{P}_R := \mathbb{F}[\mathring{Q}]. \quad (47)$$

For $\lambda \in \mathring{Q}$, let λ_+ be the dominant element of $\mathring{W}\lambda$, and let $v(\lambda)$ be the shortest element of $\mathring{W}$ sending λ to the anti-dominant chamber. Define

$$\lambda \geq \eta \iff \begin{cases} \lambda_+ > \eta_+, & \text{in strict dominance order,} \\ \lambda_+ = \eta_+ \text{ and } v(\lambda) \leq v(\eta), & \text{in finite Bruhat order.} \end{cases} \quad (48)$$

Set

$$\mathring{Q}_{\leq \lambda} := \{\eta \in \mathring{Q} \mid \eta \leq \lambda\}, \quad \mathcal{P}_{\leq \lambda} := \text{span}_{\mathbb{F}}\{X_\eta \mid \eta \leq \lambda\}. \quad (49)$$

The set $\mathring{Q}_{\leq \lambda}$ is finite and lower closed.

Theorem 5.1 (*Y-triangularity*). *For every $\mu \in M$ and $\lambda \in \mathring{Q}$, there are unique coefficients $\gamma_\lambda^R(\mu) \in \mathbb{F}^\times$ and $a_{\eta\lambda}^R(\mu) \in \mathbb{F}$ such that*

$$Y_\mu X_\lambda = \gamma_\lambda^R(\mu) X_\lambda + \sum_{\eta < \lambda} a_{\eta\lambda}^R(\mu) X_\eta. \quad (50)$$

In particular, $\mathcal{P}_{\leq \lambda}$ is invariant under every Y_μ .

For anti-dominant μ , the theorem follows by expanding a reduced expression of the positive braid lift T_{λ_μ} . For a general $\mu = \mu_1 - \mu_2$, with μ_1, μ_2 anti-dominant, one uses $Y_\mu = Y_{\mu_1} Y_{\mu_2}^{-1}$; the inverse of an invertible lower triangular operator is again lower triangular. This is Macdonald's triangularity argument written in Ion's $\mathring{Q}, M, X, Y$ notation [Mac03, 4.6.4–4.6.12].

Definition 5.2. *For generic parameters, the Ion-normalized non-symmetric Macdonald polynomial $E_\lambda^R \in \mathcal{P}_R$ is the monic common eigenvector satisfying*

$$E_\lambda^R = X_\lambda + \sum_{\eta < \lambda} c_{\lambda\eta}^R X_\eta, \quad (51)$$

$$Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R, \quad \mu \in M. \quad (52)$$

Here “generic” means that the joint characters γ_η^R are pairwise distinct on the finite lower ideal under consideration and that the denominators below do not vanish.

Proposition 5.3 (Finite triangular recursion). *Choose $\nu \in M$ such that $\gamma_\eta^R(\nu), \eta \leq \lambda$, are pairwise distinct, and write*

$$Y_\nu X_\zeta = \gamma_\zeta^R(\nu) X_\zeta + \sum_{\eta < \zeta} a_{\eta\zeta}^R(\nu) X_\eta. \quad (53)$$

Then

$$c_{\lambda\eta}^R = \frac{a_{\eta\lambda}^R(\nu) + \sum_{\eta < \zeta < \lambda} c_{\lambda\zeta}^R a_{\eta\zeta}^R(\nu)}{\gamma_\lambda^R(\nu) - \gamma_\eta^R(\nu)}. \quad (54)$$

The computation proceeds from the weights nearest to λ downward, one layer at a time.

From the polynomial representation, $T_i 1 = t_i^{1/2} 1$. Hence every Y_μ sends 1 to a scalar multiple of 1, and leading-term normalization gives

$$\boxed{E_0^R = 1.} \quad (55)$$

6 Macdonald Evaluation-Duality

6.1 Spectral points

For $\lambda \in \mathring{Q}$, define an algebraic character on the dual polynomial algebra $\mathcal{P}_{R^\iota} = \mathbb{F}[\mathring{Q}_X^\iota]$ by

$$\mathrm{sp}_R(\lambda) : \mathcal{P}_{R^\iota} \longrightarrow \mathbb{F}, \quad \boxed{X_{\mu/\sqrt{p}}^\iota(\mathrm{sp}_R(\lambda)) := \gamma_\lambda^R(-\mu)}, \quad \mu \in M. \quad (56)$$

This covers every index in $\mathring{Q}^l$, since $\mathring{Q}^l = p^{-1/2}M$. Similarly, for $\beta \in \mathring{Q}^l$, define

$$\mathrm{sp}_{R^l}(\beta) : \mathcal{P}_R \longrightarrow \mathbb{F}, \quad \boxed{X_\lambda(\mathrm{sp}_{R^l}(\beta)) := \gamma_\beta^{R^l}(-\sqrt{p}\lambda)}, \quad \lambda \in \mathring{Q}. \quad (57)$$

Equations (56)–(57) completely define the spectral points. The minus sign follows from $\omega_R(X_\lambda) = Y_{-\sqrt{p}\lambda}^l$, not from an additional choice of convention.

6.2 The duality theorem

For $f \in \mathcal{P}_R$ and $g \in \mathcal{P}_{R^l}$, define the Fourier pairing

$$[f, g]_R := (\omega_R(f)g)(\mathrm{sp}_R(0)). \quad (58)$$

The standard symmetry of this pairing is

$$[f, g]_R = [g, f]_{R^l}; \quad (59)$$

see [Mac03, 4.7.14].

Theorem 6.1 (General non-symmetric evaluation-duality). *For every $\lambda \in \mathring{Q}$ and $\beta \in \mathring{Q}^l$,*

$$\boxed{\frac{E_\lambda^R(\mathrm{sp}_{R^l}(\beta))}{E_\lambda^R(\mathrm{sp}_{R^l}(0))} = \frac{E_\beta^{R^l}(\mathrm{sp}_R(\lambda))}{E_\beta^{R^l}(\mathrm{sp}_R(0))}}. \quad (60)$$

Proof. By (46) and the eigenvalue equation (52),

$$\omega_R(f)E_\beta^{R^l} = f(\mathrm{sp}_{R^l}(\beta))E_\beta^{R^l}, \quad f \in \mathcal{P}_R. \quad (61)$$

Evaluation at $\mathrm{sp}_R(0)$ gives

$$[f, E_\beta^{R^l}]_R = f(\mathrm{sp}_{R^l}(\beta))E_\beta^{R^l}(\mathrm{sp}_R(0)). \quad (62)$$

Exchanging R and R^l gives

$$[E_\lambda^R, g]_{R^l} = g(\mathrm{sp}_R(\lambda))E_\lambda^R(\mathrm{sp}_{R^l}(0)). \quad (63)$$

Take $f = E_\lambda^R$ and $g = E_\beta^{R^l}$, and use (59). This proves (60). $\square$

7 Examples

7.1 Examples of the root-system involution

The finite simple roots of types A , B , and C are as follows:

$$\begin{array}{lll} A_{n-1} : & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n; \\ B_n : & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n, \quad \alpha_n = \varepsilon_n; \\ C_n : & \alpha_i = \varepsilon_i - \varepsilon_{i+1}, & 1 \leq i < n, \quad \alpha_n = 2\varepsilon_n. \end{array}$$

Table 1: Affine labels and dual labels.

Type	$(a_0, a_1, \dots, a_n)$	$(a_0^\vee, a_1^\vee, \dots, a_n^\vee)$
$A_n^{(1)}$	$(1, 1, \dots, 1)$	$(1, 1, \dots, 1)$
$B_n^{(1)}$	$(1, 1, 2, \dots, 2, 2)$	$(1, 1, 2, \dots, 2, 1)$
$C_n^{(1)}$	$(1, 2, \dots, 2, 1)$	$(1, 1, \dots, 1)$

Consequently,

$$A_n^{(1)} : M = \mathring{Q}, \quad (64)$$

$$B_n^{(1)} : M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\alpha_i \oplus \mathbb{Z}(2\alpha_n), \quad (65)$$

$$C_n^{(1)} : M = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i) \oplus \mathbb{Z}\alpha_n. \quad (66)$$

Now specialize to $B_n^{(1)}$. Then

$$d_1 = \dots = d_{n-1} = 1, \quad d_n = 2, \quad (67)$$

and hence

$$e_1 = \dots = e_{n-1} = 1, \quad e_n = 2. \quad (68)$$

Thus

$$A_j = \alpha_j \quad (1 \leq j < n), \quad A_n = 2\alpha_n, \quad p = e_s = 2. \quad (69)$$

By definition,

$$\alpha_i^t = \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \quad (i < n), \quad \alpha_n^t = \sqrt{2} \varepsilon_n. \quad (70)$$

Their squared lengths are

$$(\alpha_i^t, \alpha_i^t) = 1 \quad (i < n), \quad (\alpha_n^t, \alpha_n^t) = 2. \quad (71)$$

Thus the first $n - 1$ roots are short and the last root is long. The finite Cartan matrix is

$$\mathring{A}^t = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & -2 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \quad (72)$$

which is of type C. Hence

$$\boxed{(B_n^{(1)})^t = C_n^{(1)}}. \quad (73)$$

In this case, Theorem 4.2 becomes

$$\phi_{\tilde{W}}(T_j) = (T_j^l)^{-1}, \quad 1 \leq j \leq n, \quad (74)$$

$$\phi_{\tilde{W}}(Y_\mu) = X_{\mu/\sqrt{2}}^l, \quad \mu \in M, \quad (75)$$

$$\phi_{\tilde{W}}(X_\beta) = Y_{\sqrt{2}\beta}^l, \quad \beta \in \mathring{Q}, \quad (76)$$

$$\phi_{\tilde{W}}(q) = (q^l)^{-1}, \quad (77)$$

$$\phi_{\tilde{W}}(t_j) = (t_j^l)^{-1}. \quad (78)$$

Moreover,

$$\mathring{Q}^l = \frac{1}{\sqrt{2}}M = \bigoplus_{i=1}^{n-1} \mathbb{Z} \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \oplus \mathbb{Z}(\sqrt{2}\varepsilon_n), \quad (79)$$

whereas

$$M^l = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i^l) \oplus \mathbb{Z}\alpha_n^l = \sqrt{2}\mathring{Q}. \quad (80)$$

This shows precisely that the X -lattice of B_n becomes the Y -lattice of C_n , while the Y -lattice of B_n becomes the X -lattice of C_n .

7.2 The polynomial representation of type $B_n^{(1)}$

Let $n \geq 3$, and put

$$\alpha_i = \varepsilon_i - \varepsilon_{i+1} \quad (1 \leq i < n), \quad \alpha_n = \varepsilon_n, \quad (81)$$

and

$$\theta = \varepsilon_1 + \varepsilon_2, \quad \alpha_0 = \delta - \theta. \quad (82)$$

Then

$$\mathring{Q} = \mathbb{Z}^n, \quad M = \left\{ (m_1, \dots, m_n) \in \mathbb{Z}^n : \sum_{i=1}^n m_i \equiv 0 \pmod{2} \right\}. \quad (83)$$

Write

$$x_i := X_{\varepsilon_i}, \quad L := t_l^{1/2}, \quad S := t_s^{1/2}, \quad (84)$$

with

$$t_0 = t_1 = \dots = t_{n-1} = t_l, \quad t_n = t_s. \quad (85)$$

For $1 \leq i < n$, s_i swaps x_i and x_{i+1} , while s_n sends x_n to x_n^{-1} . The affine reflection satisfies

$$(s_0 f)(x_1, x_2, x_3, \dots, x_n) = f\left(\frac{1}{qx_2}, \frac{1}{qx_1}, x_3, \dots, x_n\right). \quad (86)$$

Since $X_{-\alpha_0} = qx_1x_2$, the polynomial representation is

$$T_i f = Ls_i f + (L - L^{-1}) \frac{f - s_i f}{1 - x_{i+1}/x_i}, \quad 1 \leq i < n, \quad (87)$$

$$T_n f = Ss_n f + (S - S^{-1}) \frac{f - s_n f}{1 - x_n^{-1}}, \quad (88)$$

$$T_0 f = Ls_0 f + (L - L^{-1}) \frac{f - s_0 f}{1 - qx_1x_2}. \quad (89)$$

Call $\mu \in M$ *anti-dominant* if

$$(\mu, A_i^\vee) \leq 0, \quad 1 \leq i \leq n. \quad (90)$$

For anti-dominant μ , Ion defines

$$Y_\mu = T_{\lambda_\mu}. \quad (91)$$

For example,

$$Y_{-\theta} = T_{s_\theta} T_0, \quad (92)$$

because $\lambda_\theta = s_0 s_\theta$ and $\lambda_{-\theta} = s_\theta s_0$.

Lemma 7.1. *For B_n , the $\mathbb{Z}$ -span of the $\mathring{W}$ -orbit of $\theta = \varepsilon_1 + \varepsilon_2$ is M .*

Proof. The orbit consists of the long roots $\pm \varepsilon_i \pm \varepsilon_j$, and their integer span is precisely the even-sum lattice M . $\square$

Consequently, for any $\mu \in M$, one can choose

$$\mu = \sum_{k=1}^N m_k w_k(\theta), \quad m_k \in \mathbb{Z}, \quad w_k \in \mathring{W}. \quad (93)$$

Since translations commute,

$$\lambda_\mu = \prod_{k=1}^N (w_k s_0 s_\theta w_k^{-1})^{m_k}. \quad (94)$$

Every w_k, w_k^{-1} , and s_θ can be expressed using $s_1, \dots, s_n$. Thus this is an expression for λ_μ in the affine Weyl group using $s_0, s_1, \dots, s_n$.

Remark 7.2 (Reduced expressions are essential). *The preceding identity is an identity in the affine Weyl group. A non-reduced expression cannot in general be lifted by simply replacing every s_i with T_i , because $s_i^2 = 1$ whereas*

$$T_i^2 = (t_i^{1/2} - t_i^{-1/2})T_i + 1. \quad (95)$$

For anti-dominant μ , one must first choose a reduced expression for λ_μ and then take its positive braid lift. For general μ , one may write $\mu = \mu_1 - \mu_2$ with μ_1, μ_2 anti-dominant and use $Y_\mu = Y_{\mu_1} Y_{\mu_2}^{-1}$.

7.3 Complete computation of $E_{\varepsilon_1}^{B_n^{(1)}}$

Lemma 7.3 (Triangular support of ε_1). *With respect to (48),*

$$\{\eta \in \mathring{Q} : \eta \leq \varepsilon_1\} = \{\varepsilon_1, 0\}. \quad (96)$$

Consequently,

$$\mathcal{P}_{\leq \varepsilon_1} = \text{span}_{\mathbb{F}}\{x_1, 1\}. \quad (97)$$

Proof. If $\eta_+ \leq \varepsilon_1$ in dominance order, the minimality of ε_1 among nonzero dominant elements of $\mathring{Q} = \mathbb{Z}^n$ forces $\eta_+ = 0$, hence $\eta = 0$. It remains to inspect $\mathring{W}_{\varepsilon_1} = \{\pm \varepsilon_i : 1 \leq i \leq n\}$. For the shortest element $v(\eta)$ sending η to the anti-dominant weight $-\varepsilon_1$,

$$\ell(v(\varepsilon_i)) = 2n - i, \quad \ell(v(-\varepsilon_i)) = i - 1. \quad (98)$$

Thus $\ell(v(\varepsilon_1)) = 2n - 1$ is the unique maximum. Since $u \leq w$ in Bruhat order implies $\ell(u) \leq \ell(w)$, the condition $v(\varepsilon_1) \leq v(\eta)$ is possible only when $\eta = \varepsilon_1$. $\square$

It follows that

$$E_{\varepsilon_1}^{B_n^{(1)}} = E_{\varepsilon_1}^I = x_1 + c \quad (99)$$

contains a single unknown coefficient.

Take the anti-dominant translation indexed by $-\theta = -\varepsilon_1 - \varepsilon_2$. Put

$$r_2 = s_2 s_3 \cdots s_{n-1} s_n s_{n-1} \cdots s_3 s_2. \quad (100)$$

Since $r_2(\alpha_1) = \theta$,

$$s_\theta = r_2 s_1 r_2. \quad (101)$$

Define

$$R_2 = T_2 T_3 \cdots T_{n-1} T_n T_{n-1} \cdots T_3 T_2. \quad (102)$$

Then Ion's formula gives the reduced positive braid lift

$$Y_{-\theta} = R_2 T_1 R_2 T_0. \quad (103)$$

From (86),

$$s_0 x_1 = q^{-1} x_2^{-1}, \quad \frac{x_1 - q^{-1} x_2^{-1}}{1 - q x_1 x_2} = -q^{-1} x_2^{-1}. \quad (104)$$

Therefore

$$\begin{aligned} T_0 x_1 &= L q^{-1} x_2^{-1} + (L - L^{-1})(-q^{-1} x_2^{-1}) \\ &= q^{-1} L^{-1} x_2^{-1}. \end{aligned} \quad (105)$$

Acting from right to left in (103) gives

$$R_2 T_0 x_1 = q^{-1} L^{-(2n-3)} S^{-1} x_2 + q^{-1} L^{-1} (S^{-1} - S), \quad (106)$$

$$T_1 R_2 T_0 x_1 = q^{-1} L^{-(2n-2)} S^{-1} x_1 + q^{-1} (S^{-1} - S). \quad (107)$$

In R_2 , the long-root generators occur $2n - 4$ times and the short-root generator occurs once. All the corresponding reflections fix x_1 , so

$$R_2 x_1 = L^{2n-4} S x_1, \quad R_2(1) = L^{2n-4} S. \quad (108)$$

Substitution into (107) gives

$$\begin{aligned} Y_{-\theta} x_1 &= q^{-1} L^{-(2n-2)} S^{-1} (L^{2n-4} S) x_1 + q^{-1} (S^{-1} - S) L^{2n-4} S \\ &= q^{-1} L^{-2} x_1 + q^{-1} L^{2n-4} (1 - S^2). \end{aligned} \quad (109)$$

Hence

$$\boxed{Y_{-\theta} x_1 = q^{-1} t_l^{-1} x_1 - q^{-1} t_l^{n-2} (t_s - 1).} \quad (110)$$

Each T_i satisfies $T_i 1 = t_i^{1/2} 1$. In (103), the long-root generators occur $4n - 6$ times in total and the short-root generators occur twice. Thus

$$\boxed{Y_{-\theta} 1 = t_l^{2n-3} t_s.} \quad (111)$$

Define

$$a_B := q^{-1} t_l^{-1}, \quad b_B := -q^{-1} t_l^{n-2} (t_s - 1), \quad d_B := t_l^{2n-3} t_s. \quad (112)$$

Then

$$\gamma_{\varepsilon_1}^{B_n^{(1)}}(-\theta) = a_B, \quad \gamma_0^{B_n^{(1)}}(-\theta) = d_B. \quad (113)$$

Assume $a_B \neq d_B$. Inserting (99) into the eigenvalue equation and comparing constants gives

$$b_B + c d_B = c a_B. \quad (114)$$

Therefore

$$\boxed{c_B := c = \frac{q^{-1} t_l^{n-2} (t_s - 1)}{t_l^{2n-3} t_s - q^{-1} t_l^{-1}.} \quad (115)$$

We have obtained

$$\boxed{E_{\varepsilon_1}^{B_n^{(1)}} = E_{\varepsilon_1}^I = x_1 + \frac{q^{-1} t_l^{n-2} (t_s - 1)}{t_l^{2n-3} t_s - q^{-1} t_l^{-1}.} \quad (116)$$

Proposition 7.4 (Common eigenvector property). *For generic parameters, the polynomial in (116) is a common eigenvector of the full commuting Y -family.*

Proof. By Lemma 7.3 and Theorem 5.1,

$$V_{\varepsilon_1} := \mathcal{P}_{\leq \varepsilon_1} = \text{span}_{\mathbb{F}}\{x_1, 1\} \quad (117)$$

is invariant under every Y_μ . In the ordered basis $\{x_1, 1\}$,

$$Y_{-\theta}|_{V_{\varepsilon_1}} = \begin{pmatrix} a_B & 0 \\ b_B & d_B \end{pmatrix}. \quad (118)$$

When $a_B \neq d_B$, its a_B -eigenspace is one-dimensional and is spanned by

$$E = x_1 + \frac{b_B}{a_B - d_B} = x_1 + c_B. \quad (119)$$

For $\mu \in M$, the restriction $Y_\mu|_{V_{\varepsilon_1}}$ commutes with $Y_{-\theta}|_{V_{\varepsilon_1}}$. Hence

$$Y_{-\theta}(Y_\mu E) = Y_\mu(Y_{-\theta} E) = a_B Y_\mu E. \quad (120)$$

Thus $Y_\mu E$ lies in the one-dimensional a_B -eigenspace, so there is a scalar $\gamma_{\varepsilon_1}^{B_n^{(1)}}(\mu)$ such that

$$Y_\mu E = \gamma_{\varepsilon_1}^{B_n^{(1)}}(\mu) E. \quad (121)$$

The leading term is x_1 , so triangular uniqueness identifies E with $E_{\varepsilon_1}^{B_n^{(1)}}$. $\square$

7.4 The corresponding polynomial of type $C_n^{(1)}$

Set

$$\eta_i := \frac{\varepsilon_i}{\sqrt{2}}, \quad \beta_0 := \frac{\theta}{\sqrt{2}} = \eta_1 + \eta_2 \in \mathring{Q}^\iota, \quad y_i := X_{\eta_i}^\iota. \quad (122)$$

Although an individual y_i does not belong to $\mathbb{F}[\mathring{Q}^\iota]$, every monomial index that actually occurs below lies in $\mathring{Q}^\iota$. The embedding into $\mathbb{F}[y_1^{\pm 1}, \dots, y_n^{\pm 1}]$ is only a computational notation.

The highest long root of the scaled C_n system is

$$\theta^\iota = 2\eta_1 = \sqrt{2} \varepsilon_1, \quad (123)$$

and $-2\eta_1 \in M^\iota$ is anti-dominant. The finite reflection has the reduced expression

$$s_{2\eta_1} = s_1 s_2 \cdots s_{n-1} s_n s_{n-1} \cdots s_2 s_1. \quad (124)$$

Consequently, Ion's Remark 1.4 gives

$$Y_{-2\eta_1}^{C_n^{(1)}} = T_1 T_2 \cdots T_{n-1} T_n T_{n-1} \cdots T_2 T_1 T_0. \quad (125)$$

Under the parameter-preserving map ω_R , the short-root parameter of the dual C_n system is t_l , while its long-root parameter is t_s . Thus

$$L = t_l^{1/2} = (t_{\text{sh}}^C)^{1/2}, \quad S = t_s^{1/2} = (t_{\text{lg}}^C)^{1/2}. \quad (126)$$

The affine simple root is

$$\alpha_0^\iota = \delta - 2\eta_1, \quad (127)$$

so

$$s_0(y_1) = q^{-1} y_1^{-1}, \quad X_{-\alpha_0^\iota}^\iota = q y_1^2. \quad (128)$$

Using

$$T_i X_\lambda = t_i^{1/2} X_{s_i \lambda} + (t_i^{1/2} - t_i^{-1/2}) \frac{X_\lambda - X_{s_i \lambda}}{1 - X_{-\alpha_i}}, \quad (129)$$

and acting from the rightmost end of (125), set

$$U := T_2 T_3 \cdots T_{n-1} T_n T_{n-1} \cdots T_3 T_2. \quad (130)$$

The intermediate steps are

$$T_0(y_1 y_2) = q^{-1} S^{-1} y_1^{-1} y_2, \quad (131)$$

$$T_1 T_0(y_1 y_2) = q^{-1} S^{-1} (L^{-1} y_1 y_2^{-1} + L^{-1} - L), \quad (132)$$

$$U T_1 T_0(y_1 y_2) = q^{-1} (L^{-(2n-3)} S^{-2} y_1 y_2 + L^{2n-5} - L^{2n-3}), \quad (133)$$

$$T_1 U T_1 T_0(y_1 y_2) = q^{-1} (L^{-(2n-4)} S^{-2} y_1 y_2 + L^{2n-4} - L^{2n-2}). \quad (134)$$

The last line is $Y_{-2\eta_1}^{C_n^{(1)}} X_{\beta_0}^\iota$. Therefore

$$Y_{-2\eta_1}^{C_n^{(1)}} X_{\beta_0}^\iota = q^{-1} t_l^{-(n-2)} t_s^{-1} X_{\beta_0}^\iota + q^{-1} t_l^{n-2} (1 - t_l). \quad (135)$$

In (125), the short-root generators occur $2n - 2$ times, while the long-root generators T_0, T_n occur twice. Hence

$$Y_{-2\eta_1}^{C_n^{(1)}} 1 = t_l^{n-1} t_s. \quad (136)$$

The weight β_0 is the minimal nonzero dominant element of $\dot{Q}^\iota$. The same triangular-support argument therefore gives

$$E_{\beta_0}^{C_n^{(1)}} = X_{\beta_0}^\iota + c_C. \quad (137)$$

Set

$$a_C := q^{-1} t_l^{-(n-2)} t_s^{-1}, \quad d_C := t_l^{n-1} t_s. \quad (138)$$

Equations (135)–(136) say

$$\gamma_{\beta_0}^{C_n^{(1)}}(-2\eta_1) = a_C, \quad \gamma_0^{C_n^{(1)}}(-2\eta_1) = d_C. \quad (139)$$

Solving the one-dimensional eigenvector equation yields

$$E_{\beta_0}^{C_n^{(1)}} = X_{\beta_0}^\iota + c_C, \quad c_C = \frac{q^{-1} t_l^{n-2} (t_l - 1)}{t_l^{n-1} t_s - q^{-1} t_l^{-(n-2)} t_s^{-1}}. \quad (140)$$

7.5 An explicit B_n – C_n evaluation-duality identity

Take

$$\lambda = \varepsilon_1, \quad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \quad (141)$$

Since $\beta_0 = \theta/\sqrt{2}$, the definitions of the spectral points give

$$X_{\beta_0}^\iota(\text{sp}_R(\varepsilon_1)) = \gamma_{\varepsilon_1}^R(-\theta) = a_B, \quad (142)$$

$$X_{\beta_0}^\iota(\text{sp}_R(0)) = \gamma_0^R(-\theta) = d_B. \quad (143)$$

On the other side, $-\sqrt{2}\varepsilon_1 = -2\eta_1 = -\theta^l$, so

$$x_1(\mathrm{sp}_{R^l}(\beta_0)) = \gamma_{\beta_0}^{R^l}(-\theta^l) = a_C, \quad (144)$$

$$x_1(\mathrm{sp}_{R^l}(0)) = \gamma_0^{R^l}(-\theta^l) = d_C. \quad (145)$$

Here $x_1 = X_{\varepsilon_1}$ is the B_n -side coordinate; it should not be confused with the formal dual-side symbols y_i .

Consequently,

$$\mathcal{E}_{\varepsilon_1}^R(\mathrm{sp}_{R^l}(\beta_0)) = \frac{a_C + c_B}{d_C + c_B}, \quad (146)$$

whereas

$$\mathcal{E}_{\beta_0}^{R^l}(\mathrm{sp}_R(\varepsilon_1)) = \frac{a_B + c_C}{d_B + c_C}. \quad (147)$$

Thus general non-symmetric evaluation-duality becomes the explicit identity

$$\boxed{\frac{a_C + c_B}{d_C + c_B} = \frac{a_B + c_C}{d_B + c_C}}. \quad (148)$$

Substituting (112), (115), (138), and (140) gives

$$\frac{q^{-1}t_l^{-(n-2)}t_s^{-1} + c_B}{t_l^{n-1}t_s + c_B} = \frac{q^{-1}t_l^{-1} + c_C}{t_l^{2n-3}t_s + c_C}. \quad (149)$$

Both sides simplify to

$$\boxed{\frac{q^{-1}t_l^{n-2}(t_l + t_s - 1) - q^{-2}t_l^{-(n-1)}t_s^{-1}}{t_l^{3n-4}t_s^2 - q^{-1}t_l^{n-2}}}. \quad (150)$$

This explicitly verifies normalized evaluation-duality for the nontrivial pair

$$\lambda = \varepsilon_1, \quad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \quad (151)$$

8 Guiding Principle

The two commutative parts of DAHA have different interpretations:

$$X_\beta \quad \text{acts as a coordinate or multiplication operator,} \quad (152)$$

whereas

$$Y_\mu \quad \text{acts as a spectral or } q\text{-difference operator.} \quad (153)$$

The DAHA involution exchanges these roles:

$$\boxed{\text{coordinate variables} \longleftrightarrow \text{spectral variables.}} \quad (154)$$

This exchange is the algebraic origin of Macdonald duality.

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